Thermoelectrical energy conversion

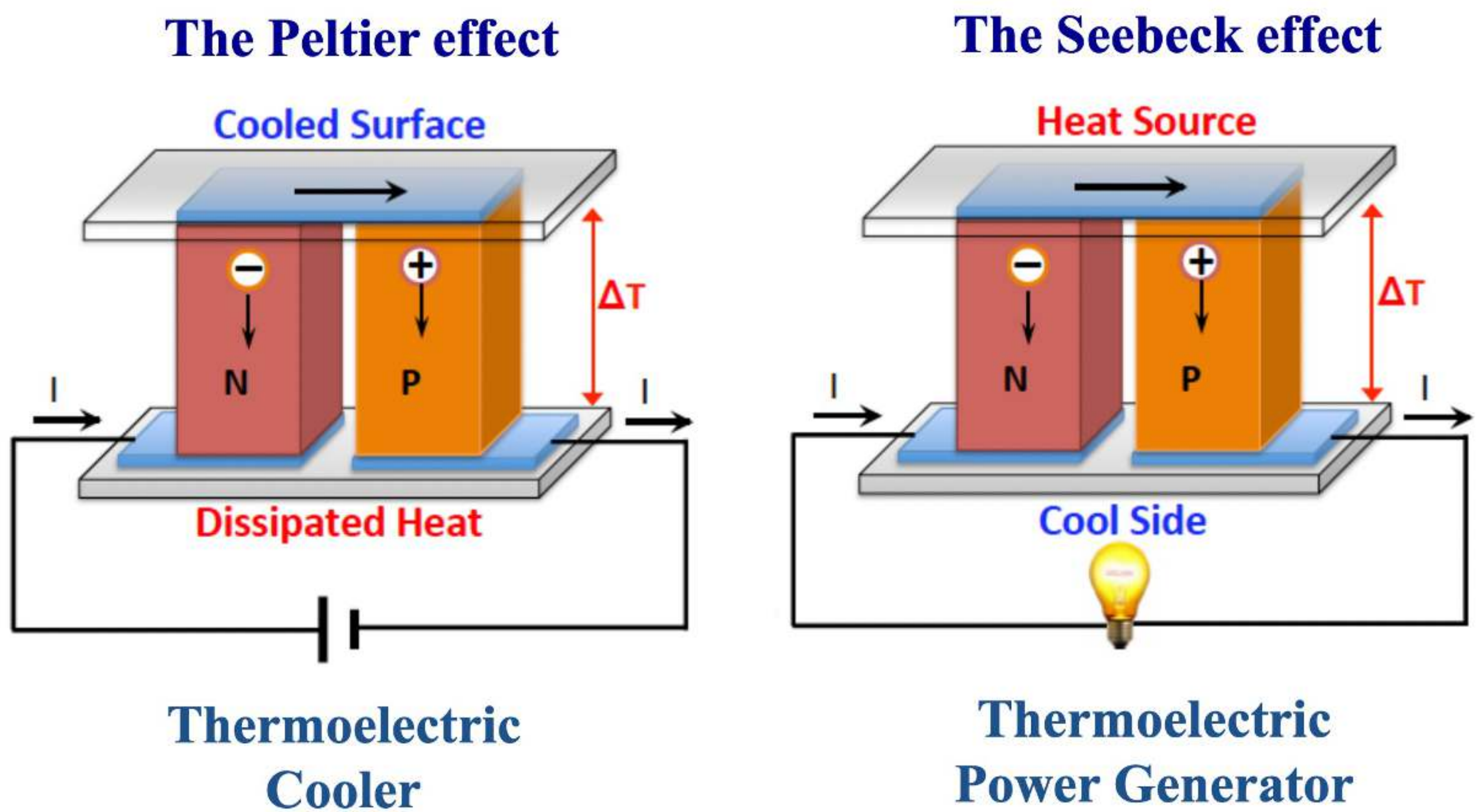
MT25-30104 - Material Konversi Energi

Pertemuan ke-5

Eka Nurfani

Thermoelectric Effect

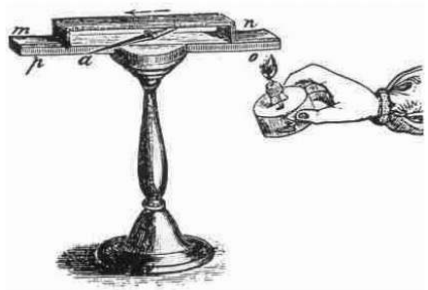
- **Thermoelectric Effect:** konversi langsung perbedaan suhu menjadi tegangan listrik dan sebaliknya



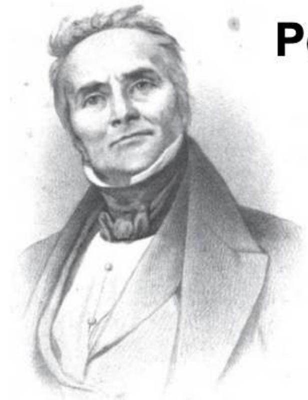
Sumber: Soonil Lee, 2018, "Introduction to Thermoelectrics"

Brief History of Thermoelectrics

Seebeck Effect (1821-3)



Thomas Johann Seebeck



Peltier Effect (1834)

In 1834, Jean Charles Athanase Peltier found that an electrical current would produce heating or cooling at the junction of two dissimilar metals.

Gustav Magnus

In 1851 Gustav Magnus discovered the Seebeck voltage does not depend on the distribution of temperature along the metals between the junctions an indication that the thermopower is a thermodynamic state function.

Basis for thermocouple

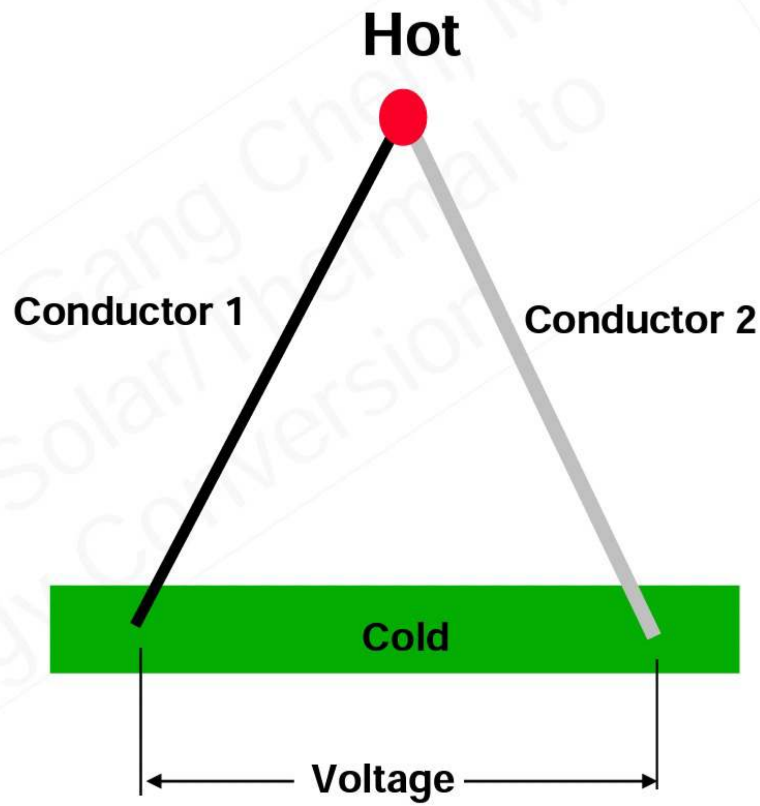
Thomson Effect (1851)

In 1851, William Thomson (later Lord Kelvin) issued a comprehensive explanation of the Seebeck and Peltier Effects and described their interrelationship (known as the Kelvin Relations).

http://www.thermoelectrics.caltech.edu/thermoelectrics/history.html

Seebeck Effect

Thomas Johann Seebeck
1770-1831

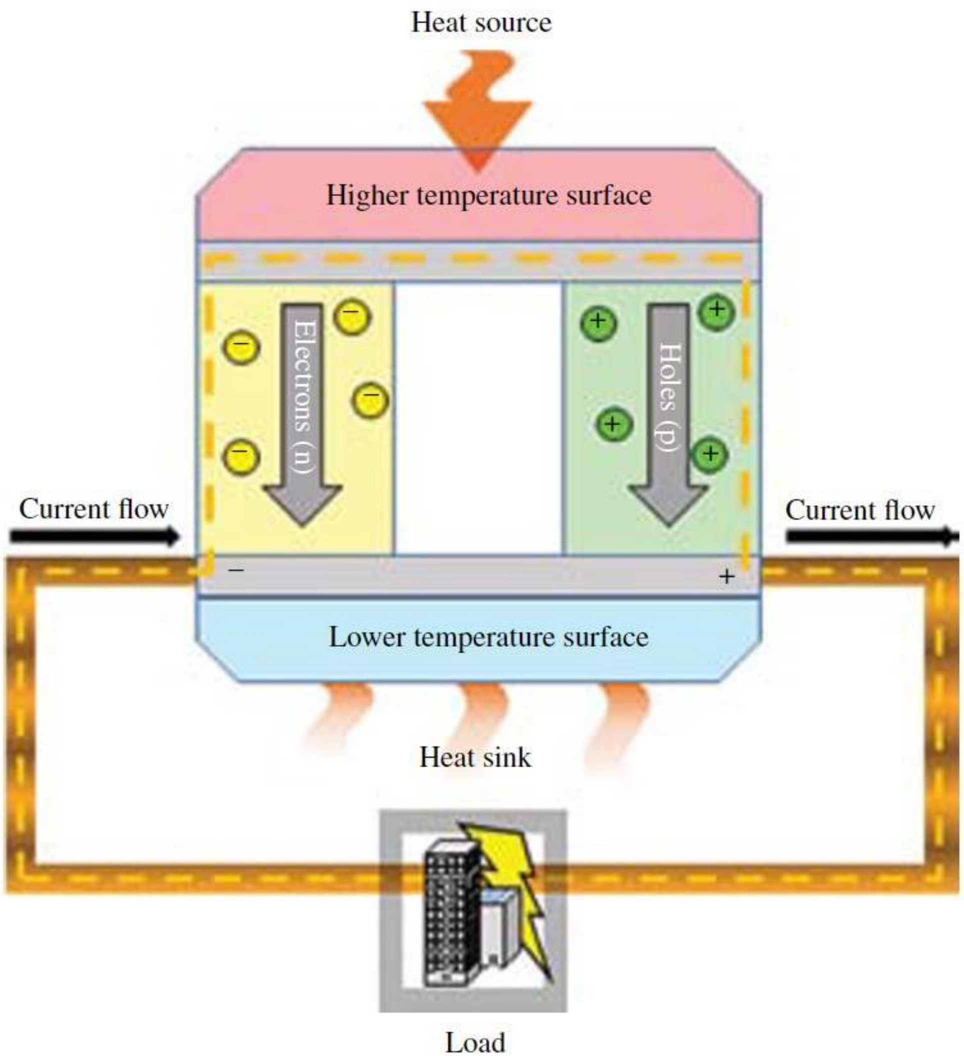


Seebeck effect: Discovered in 1821
Temperature difference generates voltage

http://www.sil.si.edu/silpublications/dibner-library-lectures/scientific-discoveries/text-lecture.htm

Sumber: Gang Cheng, slide kuliah “Introduction to Thermoelectricity”

Seebeck Effect



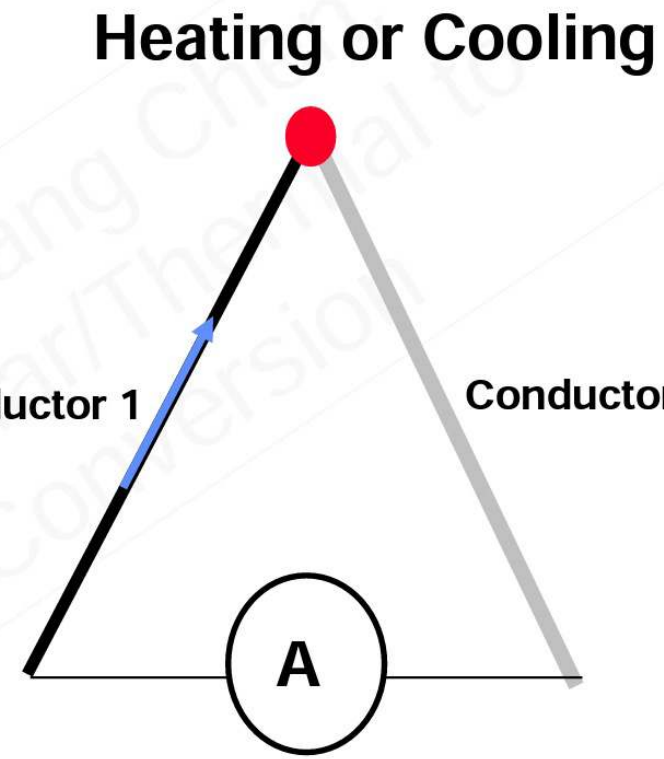
- Pembangkitan listrik melalui pertemuan dua konduktor yang berbeda

$$V = S(T_H - T_C)$$

- S = Seebeck coefficient
- *Free carrier* di ujung panas akan memiliki energi kinetik yang lebih besar dan cenderung berdifusi ke ujung dingin.

Peltier Effect

Jean Charles Athanase Peltier
1785-1845



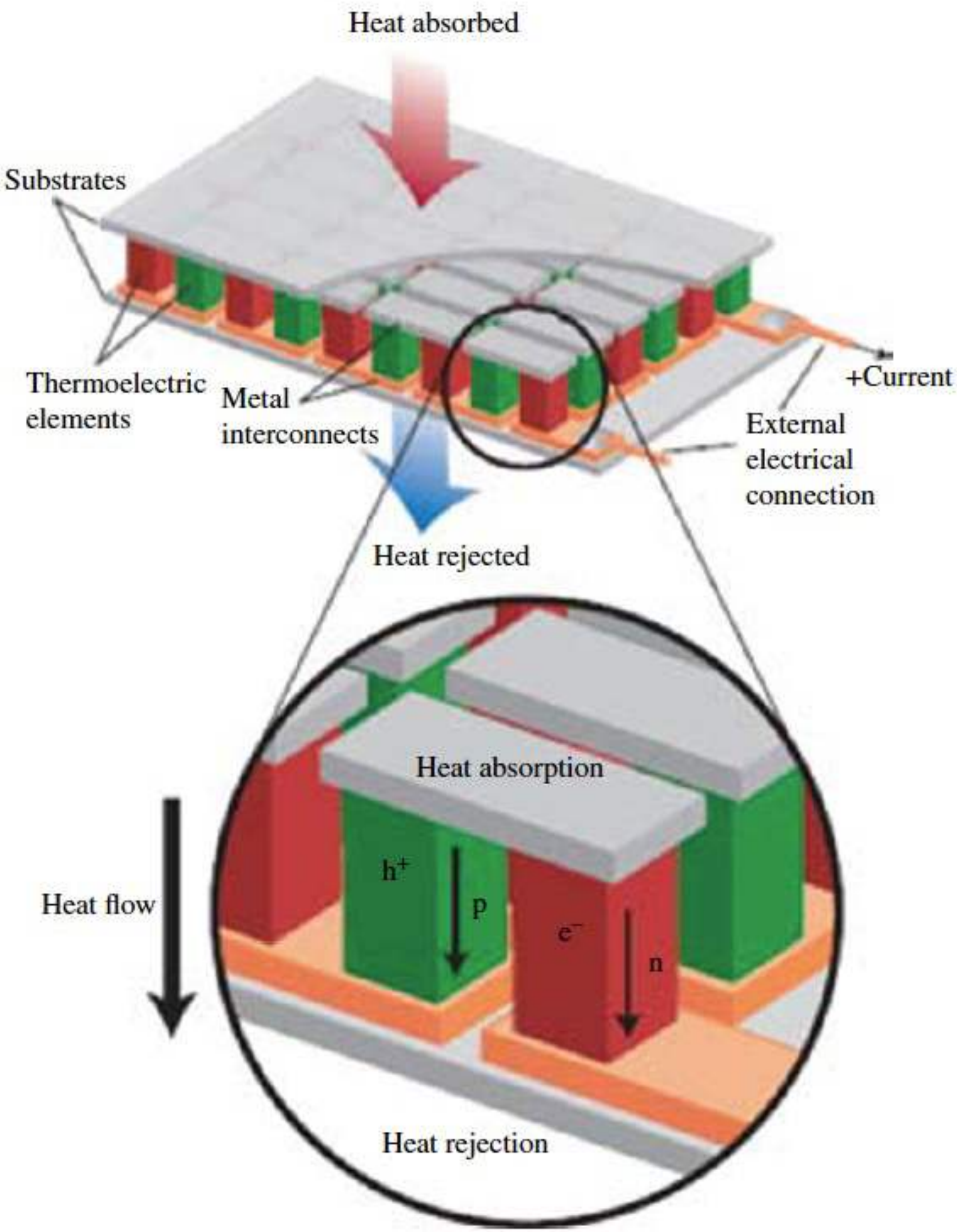
Peltier Effect: Discovered in 1834
An electrical current creates a cooling or heating effect at the junction depending on the direction of current flow.

Sumber: Gang Cheng, slide kuliah “Introduction to Thermoelectricity”

Peltier Effect

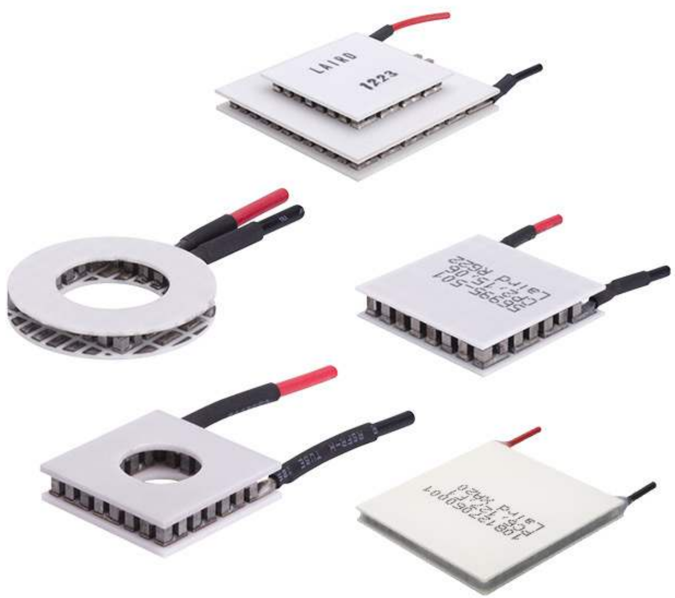
$$Q = STI$$

- I = arus
- Q = aliran panas
- Terdiri dari tipe-n dan tipe-p, dengan hubungan listrik secara seri dan termal secara paralel

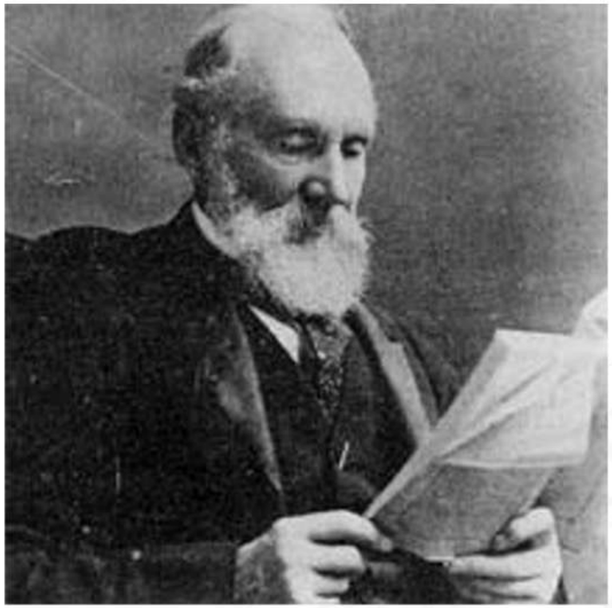


Peltier Effect

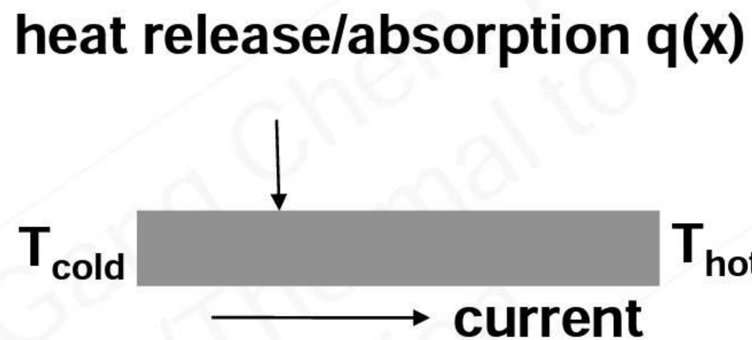
- The effect creates a temperature difference by transferring heat between two electrical junctions.
- When the current flows through the junctions of the two conductors, heat is removed at one junction.
- Heat is deposited at the other junction.



Thomson Effect



William Thomson
(Lord Kelvin)
1824 – 1907



Thomson effect predicted, 1855

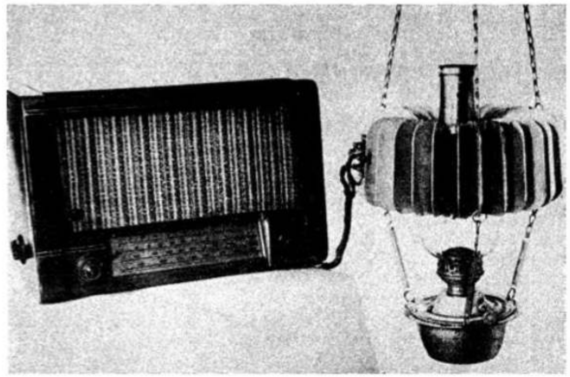
Efek Thomson menggambarkan pemanasan atau pendinginan reversibel, dalam bahan semikonduktor homogen, ketika ada aliran arus listrik dan gradien suhu.

http://www.sil.si.edu/silpublications/dibner-library-lectures/scientific-discoveries/text-lecture.htm

Sumber: Gang Cheng, slide kuliah “Introduction to Thermoelectricity”

Brief History of Thermoelectrics

Figure of Merit, ZT (1949)



In 1949 Abram Fedorovich Ioffe developed the modern theory of thermoelectricity using the concept of the 'figure of merit' zT.

One of the first demonstrations of 0 C cooling was by H. Julian Goldsmid in 1954 using thermoelements based on Bi_2Te_3, and one of the first to utilize the thermoelectric quality factor, identifying the importance of high mobility and effective mass combination and low lattice thermal conductivity in semiconductors

In 1995, Glen Slack summarized the material requirements succinctly in the "phonon-glass electron-crystal" concept.

http://www.thermoelectrics.caltech.edu/thermoelectrics/history.html

Figure of merit

- The thermoelectric figure of merit (or thermoelectric efficiency) ZT is expressed as a dimensionless parameter:

$$ZT = \frac{S^2 T \sigma}{\lambda}$$

- T = temperature mutlak
- S = Seebeck coef.
- λ = konduktifitas termal
- σ = konduktifitas listrik

Figure of merit

- To obtain maximum ZT, the material should be **semiconducting**.

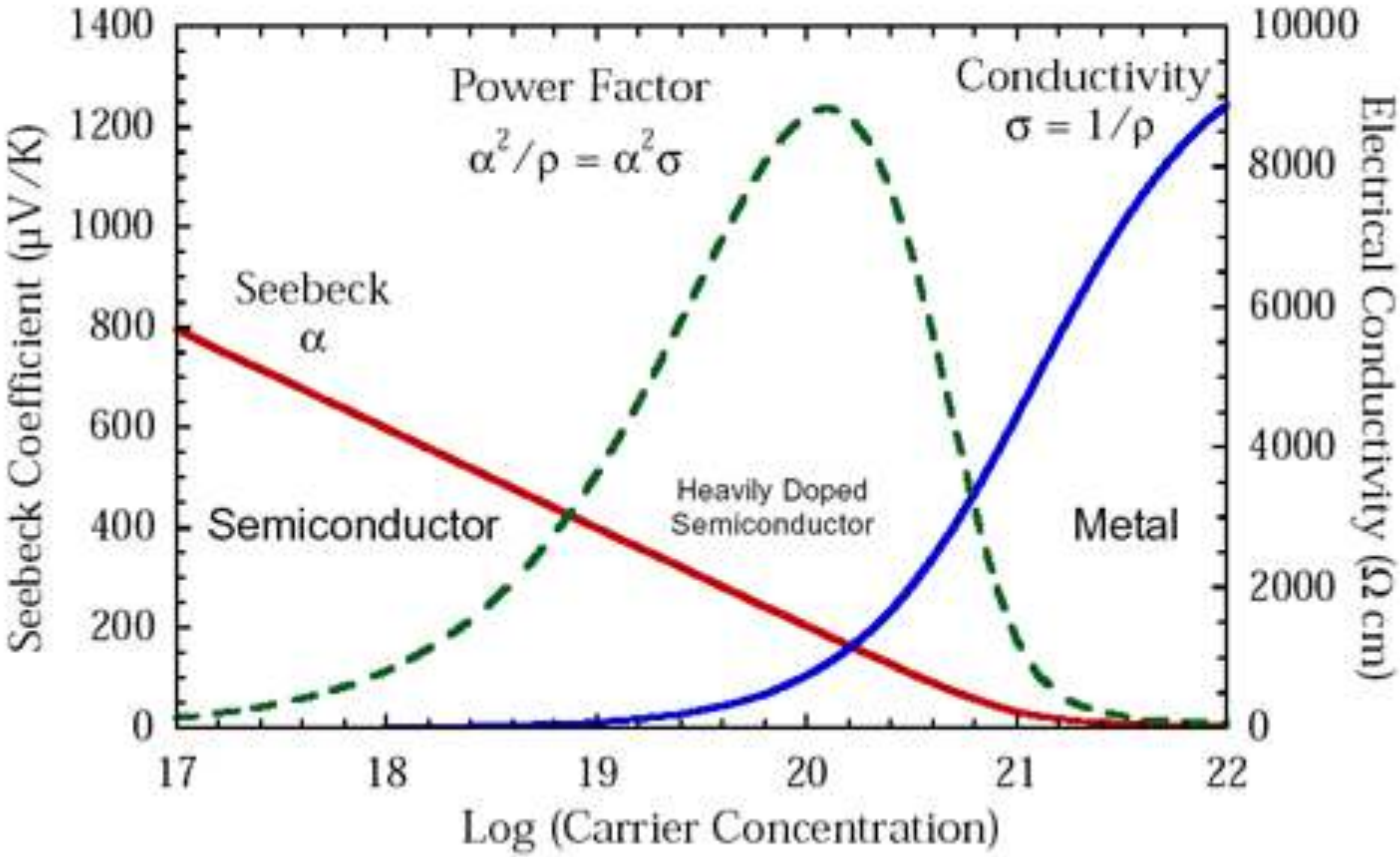


Figure of merit

- The key strategy to improve the thermal energy conversion efficiency is to **increase the Seebeck coefficient (S) and the electrical conductivity (σ)** while **reducing the thermal conductivity (λ).**

- The figure of merit reaches its maximum value around a carrier concentration of 10^{25}–10^{26} m^{-3}, which corresponds to heavily doped or near-degenerate semiconductors.

It is difficult to increase the electrical conductivity of the material by increasing the carrier concentration without affecting other parameters

Figure of merit

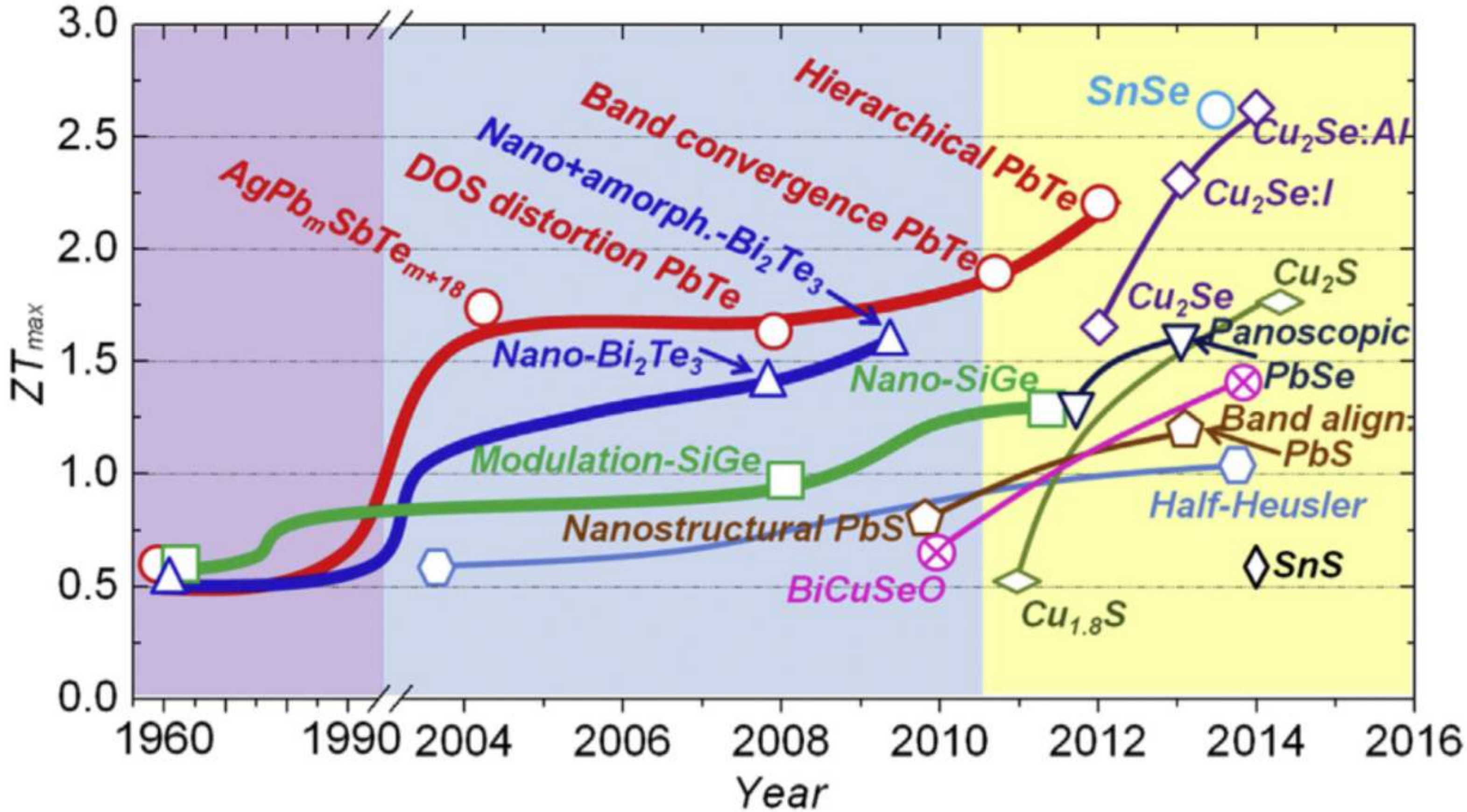
- Thermal conductivity affected by:

$$\boldsymbol{\lambda = \lambda_L + \lambda_e + \lambda_b}$$

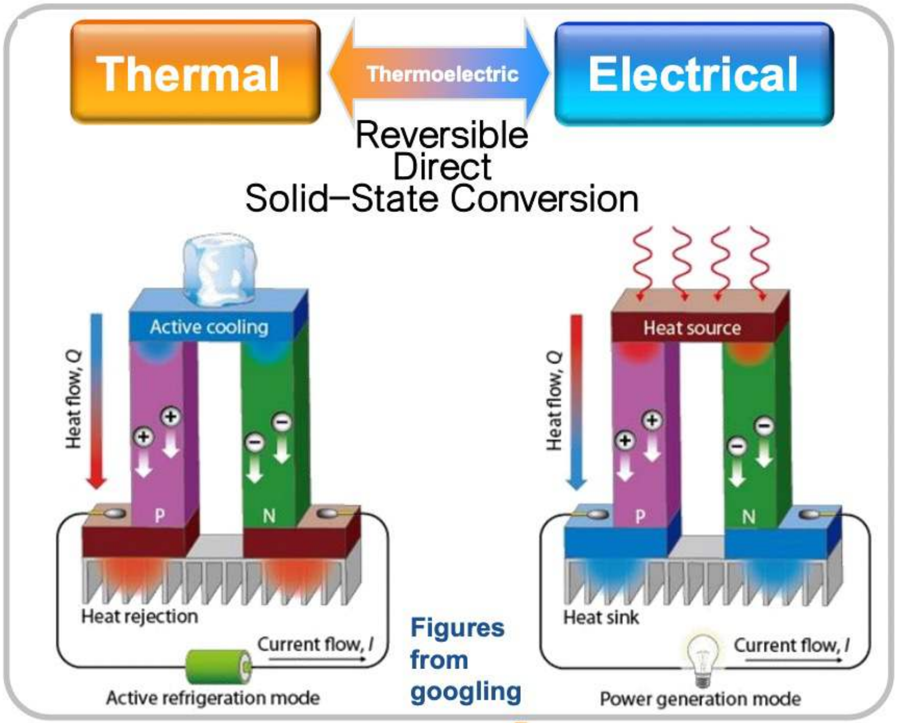
- lattice contribution, λ_L (main contribution)
- electronic or hole contribution, λ_e
- bipolar contribution, λ_b

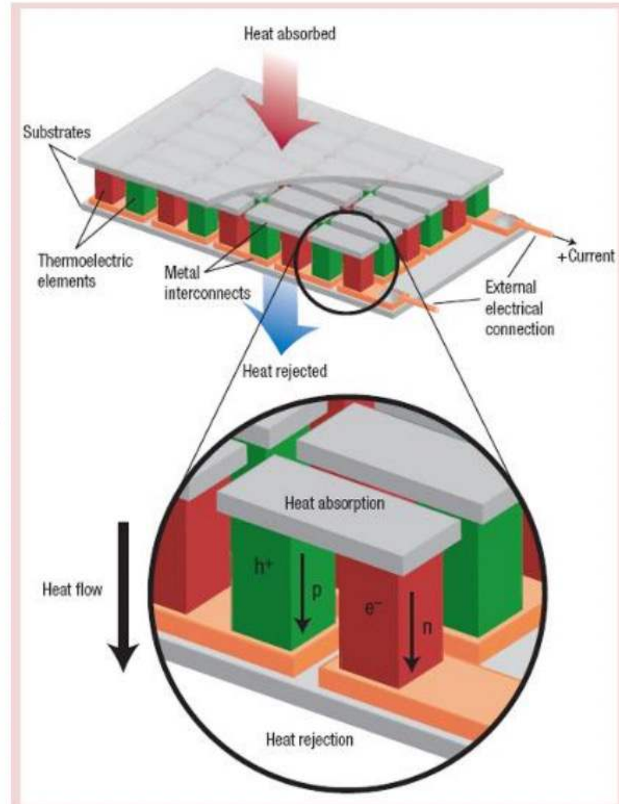
Figure of merit

- **Strategi peningkatan ZT**



Thermoelectric Applications





Snyder et al., *Nature* (2008)

Thermoelectric Cooling (Peltier Effect)

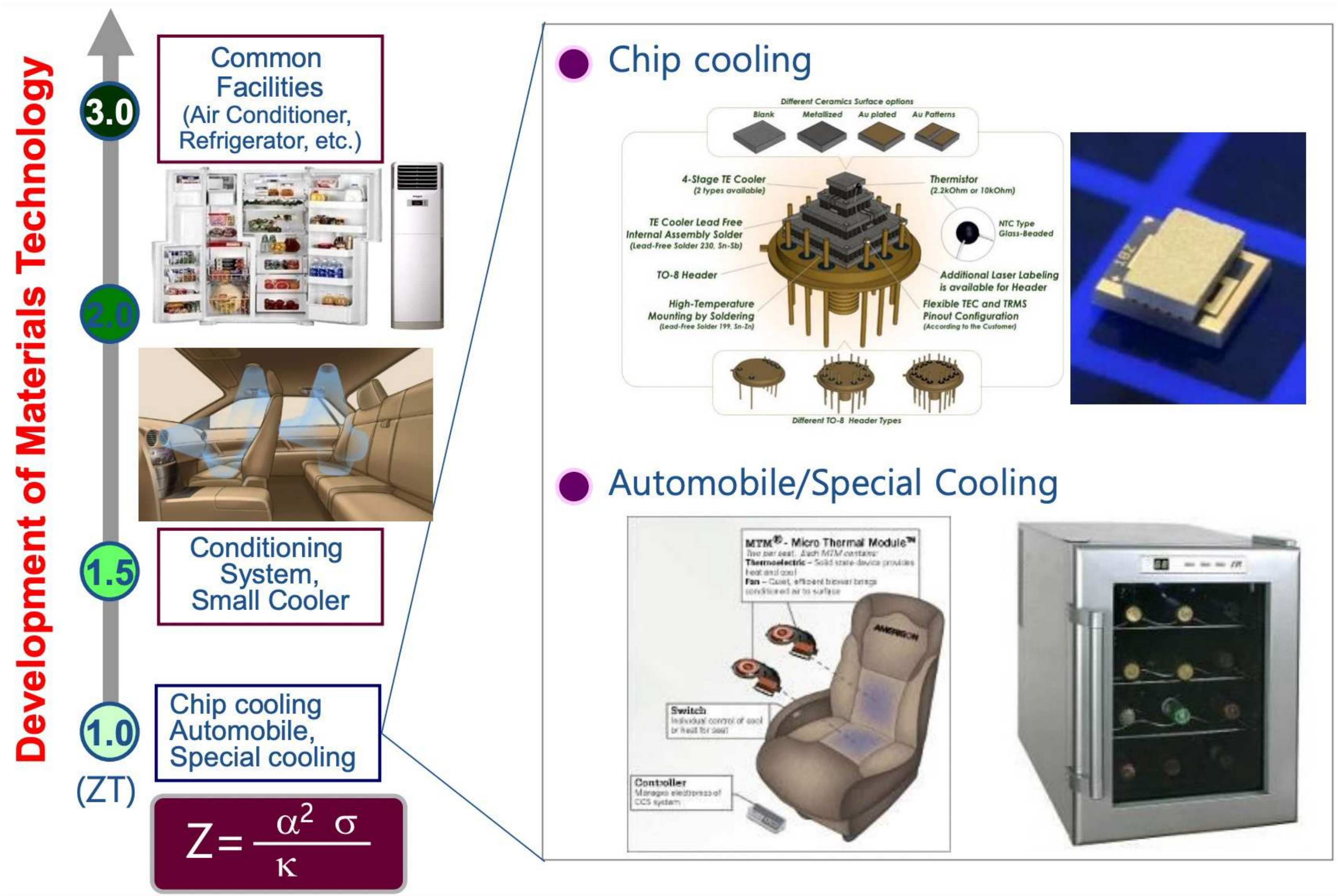
- Refrigerant–free cooling
- No–noise
- Precise temp. control
- Fast response
- Long life time
- Small local cooling

Thermoelectric Generation (Seebeck Effect)

- CO_2–free
- Recycle waste heat energy
- High reliability, No maintenance
- Stable output power (24h operation)
- DC/Micro/Independent power generation

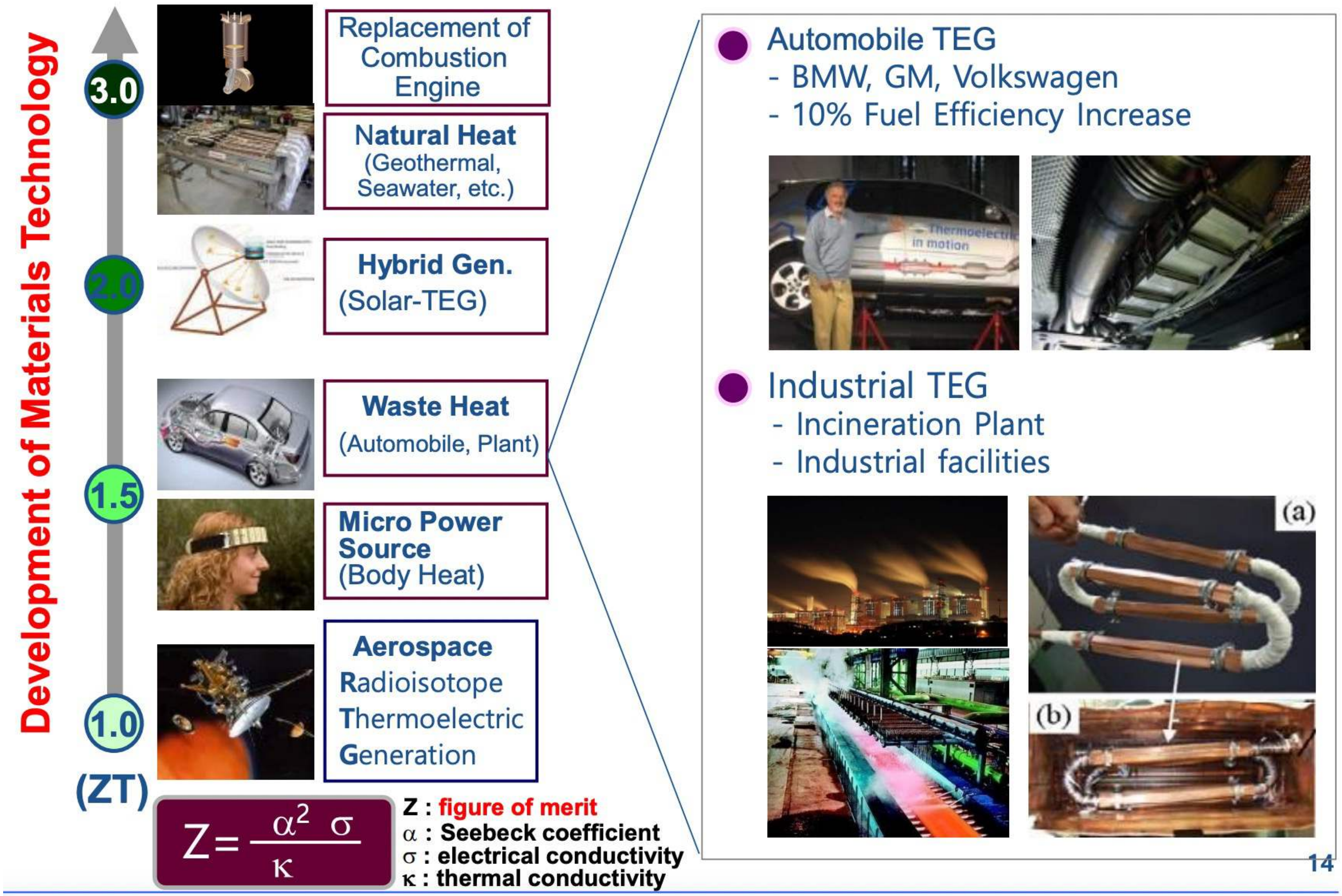
Sumber: Soonil Lee, 2018, “Introduction to Thermoelectrics”

TEC Applications



Sumber: Soonil Lee, 2018, “Introduction to Thermoelectrics”

TEG Applications using Waste Heat



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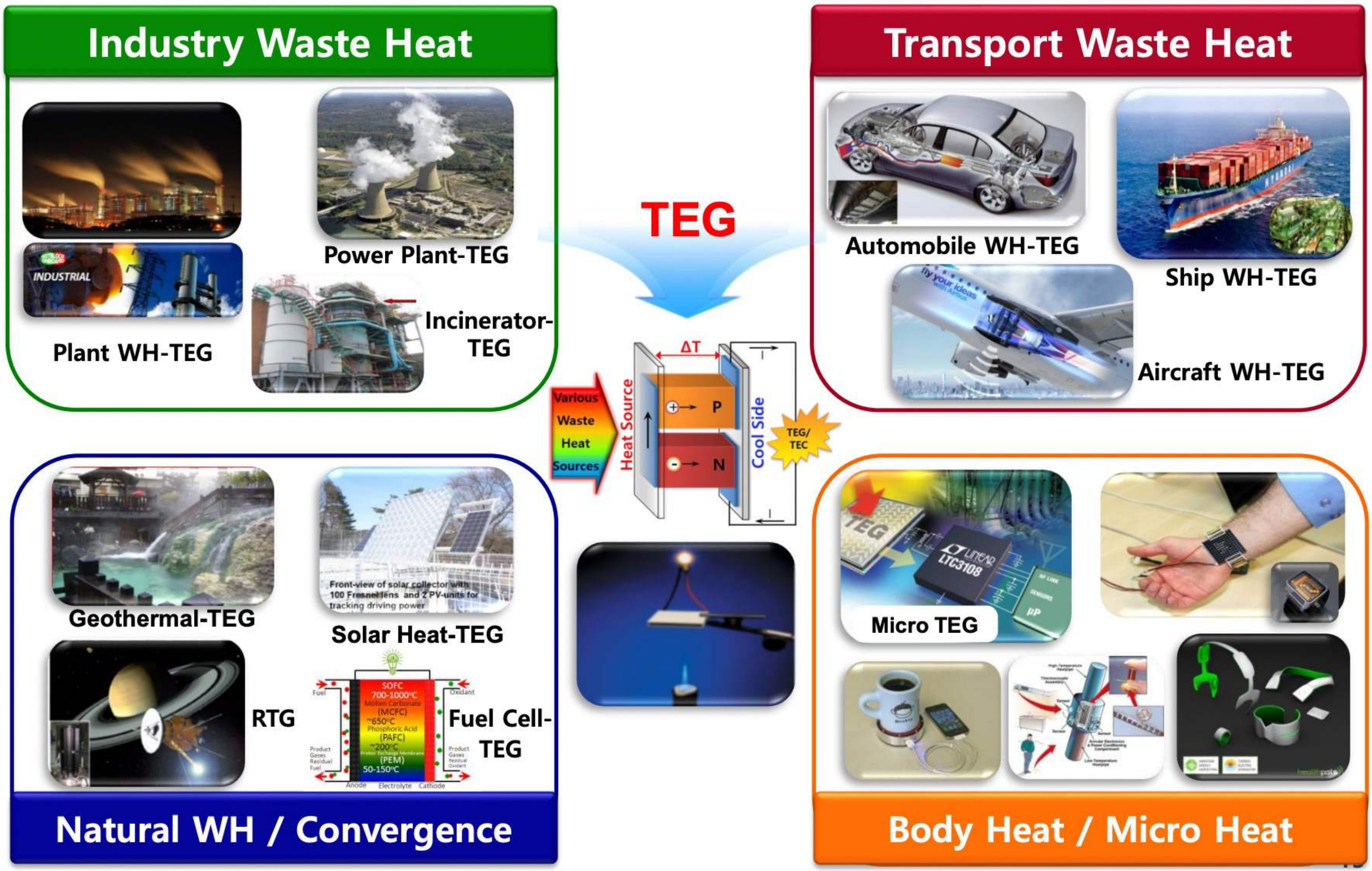
TEG Applications using Waste Heat



Sumber: Soonil Lee, 2018, “Introduction to Thermoelectrics”

TEG Applications

Portable TEG for outdoor

Sumber: Soonil Lee, 2018, “Introduction to Thermoelectrics”

Thermoelectric Materials

General category	Thermoelectric materials	ZT	Temperature (K)
Low temperature	$FeSb_2$	0.005	10
	BiSb	0.62	100
	Si nanowire arrays	~1	200
	$CsBi_4Te_6$	0.84	225
	Si nanowires	0.6	300
	β-$K_2Bi_8Se_{13}$	0.23	300
	GaN	0.0017	300
	Bi_2Te_3	0.9	300–450
	PbTe	0.8|1–1.5	450–800
	$(AgSbTe_2)(GeTe)$	1.2	500
	Tl_9BiTe_6	1.2	500
	Tl_2SnTe_5	1	500

Moderate temperature	$Yb_{0.19}Co_4Sb_{12}$	1	600
	$CaCd_2Sb_2$	0.04	600
	$EuCd_2Sb_2$	0.60	617
	Sb-doped TiNiSn	0.45	650
	Polycrystalline $Ag_{0.08}Pb_{22}SbTe_{20}$	1.5	660
	β-Zn_4Sb_3	1.17–1.32	670
	$In_{0.2}Co_4Sb_{12}$	0.7	675
	$(Zr_{0.5}Hf_{0.5})_{0.5}Ti_{0.5}NiSn_{0.998}Sb_{0.002}$	1.4	700
	Ag_9TiTe_5	1.23	700
	$TlIn_{0.94}Yb_{0.06}Te_2$	1.8	700
	$AgPb_{18}SbTe_{20}$	1.7–2.1	700–800
	$AgSn_4SbTe_6$	1	710
	$AgSbTe_2$	1.3–1.65	570–720
	$Ce_{0.9}Fe_3CoSb_{12}$	1.1	740
	$(AgSbTe_2)_{0.2}(GeTe)_{0.8}$	1.4	750
	$(AgSbTe_2)_{0.15}(GeTe)_{0.85}$	1.5	750
	$Ag_{0.015}Pb_{0.985}Se$	1.0	770
	BiCuSeO	0.7	773
	$In_{0.1}Yb_{0.2}Co_4Sb_{12}$	0.9	782
	$NaCoO_2$ (crystal)	1	800
	PbTe with Tl	1.5	773
	$Ca_{0.18}Ni_{0.03}Co_{3.97}Sb_{12.4}$	1	800
	$Ba_{0.3}Co_4Sb_{12}$	0.8	800
	$Ca_{0.08}Co_4Sb_{12.45}$	0.45	800
	$Zr_{0.5}Hf_{0.5}Ni_{0.8}Pd_{0.2}Sn_{0.99}Sb_{0.01}$	0.7	800
	$NaCo_2O_4$ (single crystal sample)	1	800

Thermoelectric Materials

High temperature	$Ba_{0.30}Ni_{0.05}Co_{3.95}Sb_{12}$	1.3	900
	$Ba_8Ga_{16}Ge_{30}$	1.35	900
	$Si_{33}P_{13}Te_8$	0.45	900
	$Ca_{2.7}Gd_{0.3}Co_4O_{9+\delta}$	0.24	973
	$Al_{0.4}Ga_{0.6}N$	0.2	1000
	$Ca_3Co_4O_9$ (crystal)	0.9	1000
	$NaCo_2O_4$ (polycrystalline sample)	0.8	1000
	$Bi_2Sr_2Co_2O_y$ (crystal)	1.14	1000
	Nb-doped $SrTiO_3$	0.37	1000
	La-doped $SrTiO_3$	0.26	1000
	$Al_{0.02}Zn_{0.98}O$	0.3	1000
	$NaCoO_2$ (ceramic)	0.78	1000
	$Hf_{0.75}Zr_{0.25}NiSn_{0.975}Sb_{0.025}$	0.8	1025
	Ca-doped $(ZnO)_3(In_2O_3)$	0.31	1053
	$Yb_{14}MnSb_{11}$	1	1200
	$Si_{0.8}Ge_{0.2}$	0.6	1200
	Boron-based materials	0.0007	1300

Engineering Thermoelectric Materials

The maximum efficiency of a thermoelectric material for both power generation and cooling is determined by its figure of merit (*zT*):

Device: $$\eta_{max} = \frac{\Delta T}{T_h} \cdot \frac{\sqrt{1+zT}-1}{\sqrt{1+zT}+T_c/T_h}$$

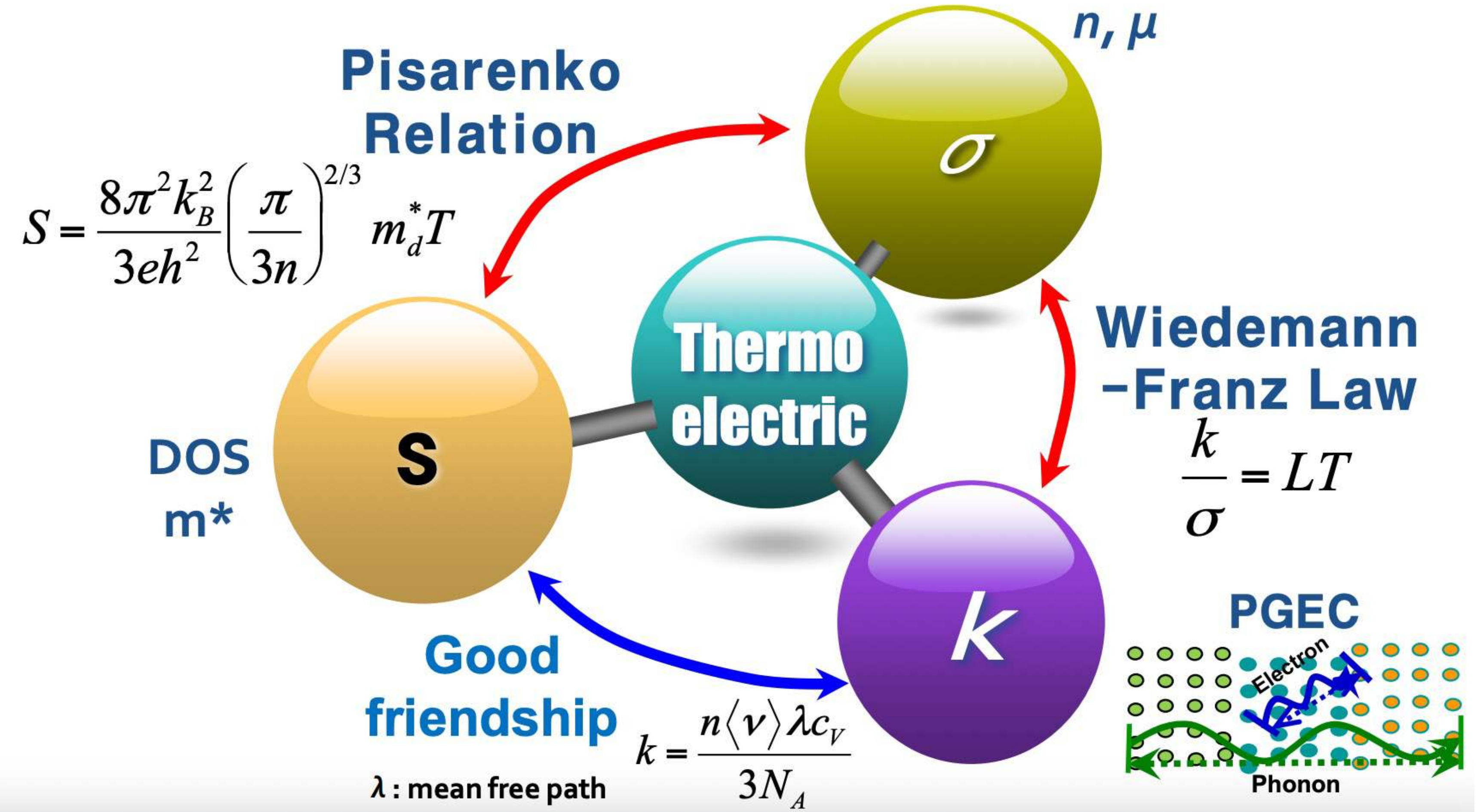
$\frac{\Delta T}{T_h}$: Carnot efficiency

Material: $$zT = \frac{\overbrace{S}^{\text{Seebeck coefficient}}{}^2\,\overbrace{\sigma}^{\text{Electrical conductivity}}}{\underbrace{k}_{\text{Thermal conductivity}}}T$$

Sumber: Soonil Lee, 2018, "Introduction to Thermoelectrics"

Engineering Thermoelectric Materials

$$zT = \frac{S^2\sigma}{k}T = S^2 n_i\left(\frac{\mu_i}{k_e + k_l}\right)qT$$



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Engineering Thermoelectric Materials

$$zT = \frac{S^2 \sigma T}{k}$$

Higher Seebeck coefficient (S)
Higher electrical conductivity (σ)
Lower thermal conductivity (k)

Engineering Decoupling

Glasses have low lattice thermal conductivity but low Seebeck coefficient.
Crystals have high electrical conductivity but high thermal conductivity.

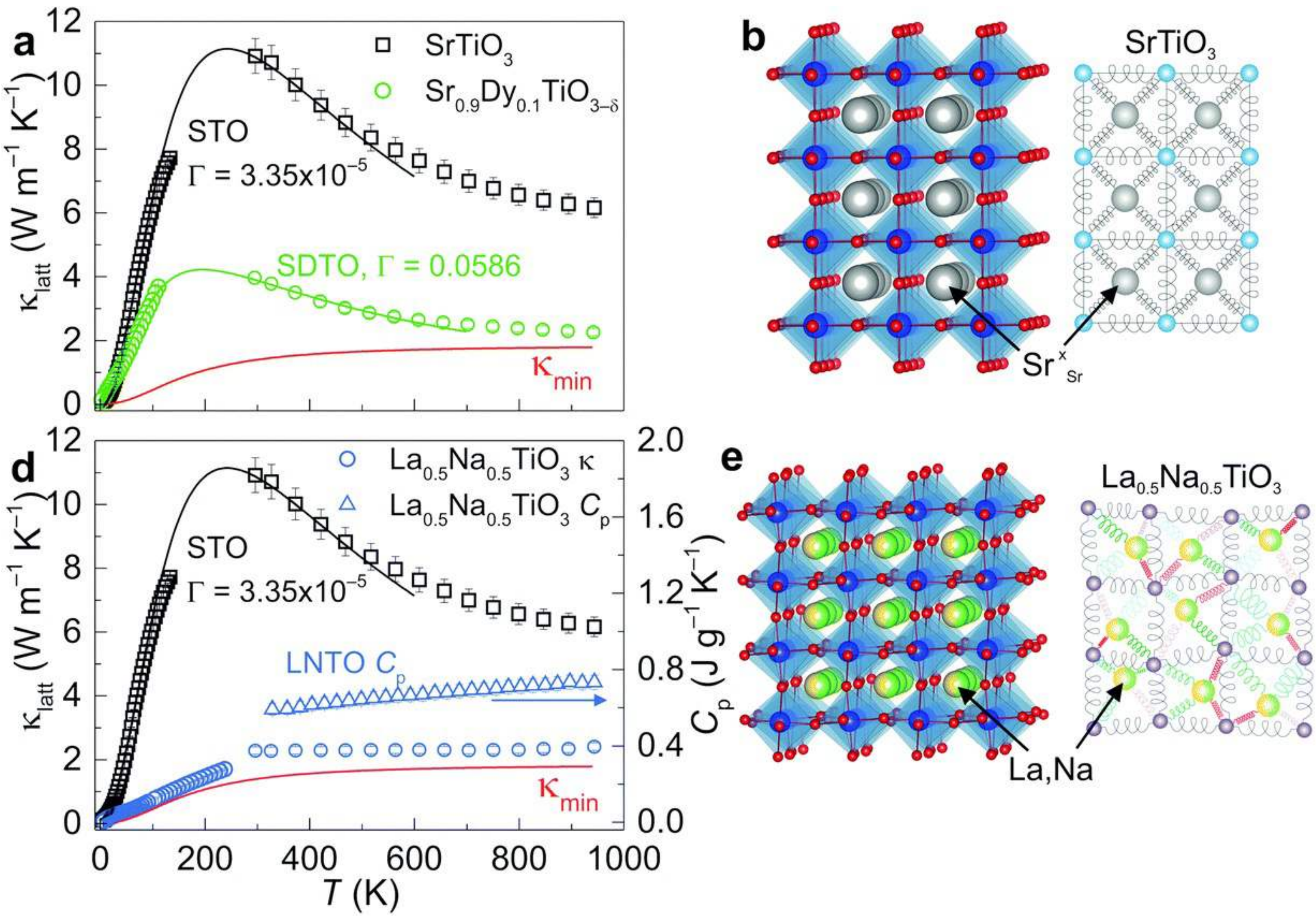
"Phonon-glass electron-crystal (PGEC)"

G. A. Slack, in CRC Handbook of Thermoelectrics (ed. Rowe, M.) 407-440 (CRC, Boca Ralon, 1995)

Original Paper:
A. F. Ioffe, S. V. Airapetiants, A. F. Ioffe,
N. V. Kolomoets and L. S. Stil'bans,
Dokl. Akad. Nauk. SSSR 106,981 (1956).

Sumber: Soonil Lee, 2018, "Introduction to Thermoelectrics"

Engineering Thermoelectric Materials



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